

PALAK DWIVEDI

[GitHub](#)

[Hugging Face](#)

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[LeetCode](#)

[Google Scholar](#)

[Portfolio Web](#)

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SKILLS

- **Languages** Python, Java, C++, HTML, CSS, JavaScript, TypeScript
- **Tools/Frameworks** Scikit-learn, TensorFlow, Keras, PyTorch, OpenCV, NumPy, Pandas
- **Databases** MySQL, PostgreSQL, MongoDB, Vector DB
- **Other** Machine Learning, DevOps, AWS, Docker, RAG, API's

EDUCATION

B. Tech, Computer Science & Engineering(AIML)

August 2024 – August 2028

Woxsen University, Hyderabad

- CGPA: 8.32

PROJECTS

Vynce — Synchronized Music Listening Platform

[Vynce](#)

Full-stack synced listening rooms (FastAPI, SQLAlchemy, WebSockets, JWT/bcrypt) with an 8M+ song catalog and a custom MLP recommender; containerized with Docker, deployed on Hugging Face Spaces.

Diabetic Retinopathy Model — Multimodal Clinical Decision Support

[Streamlit App](#)

[GitHub](#)

An advanced, end-to-end clinical decision-support AI solution designed for Diabetic Retinopathy (DR) screening. This system fuses Deep Transfer Learning (Computer Vision), Machine Learning, and Retrieval-Augmented Generation (RAG) to provide ophthalmologists with grounded, evidence-based clinical explanation syntheses.

WORK EXPERIENCE

Research Intern — BITS Pilani, Hyderabad Campus

November 2025 – January 2026

- Modeled IEEE 39-Bus, MVMicrogrid, and Wind Farm networks in DigSILENT PowerFactory; ran symmetrical/unsymmetrical fault simulations with CCT estimation.
- Applied Park's DQ Transformation for feature engineering; built SVM, Random Forest, XGBoost, and GRU models reaching 82–92% fault-classification accuracy across grid topologies.

Research & Development Intern — Robotics Lab, Woxsen University

Feburary 2026 – Present

- Paid IIT Mandi iHub& HCIFoundation (NM-ICPS funded) project; led integration of the full hardware stack (Raspberry Pi 5, Intel RealSense T265, IMU, GPS) on a bicycle prototype.
- Designed a real-time CV gradient-detection pipeline (Canny edge detection, Hough Transform, EMA smoothing) achieving $\pm 2^\circ$ pitch accuracy; trained fault-detection models (88–96% accuracy).

Freelance Advisor — INLICL

- Advise clients on financial products and solutions while managing client relationships and generating ₹60K–₹1L in monthly earnings.

CERTIFICATES

AWSCloudPractitioner Essentials — AWS Training & Certification

Technology Job Simulation (Coding & Development) — Deloitte, via Forage

DataAnalytics Job Simulation (Data Analysis & Forensic Technology) — Deloitte, via Forage